WELCOME!

Online Training Course

Foundations in Mobile Phone Data (MPD) for Policy:
Modern Data Workflows

Course overview

- 8 online training sessions:
  - Designing MPD for Policy pipelines
  - Privacy-by-design for MPD pipelines
  - Input data quality assurance and cleaning
  - Continuity models
  - Meaningful Locations and Usual Environments
  - Calculating aggregates
  - Quality Assessment of Estimates for Decision-Making
  - From Aggregates to Estimates - population inference from MPD and standard data sources
- 5 self-paced topics:
  - Recap of fundamentals *(this session)*
  - Use case examples
  - Mobile Phone Data: datasets, fields, types
  - Reference and Complementary Data
  - Considerations for Multi-MNO data pipelines

Recap of fundamentals of MPD, data pipelines, data products, data types

Module 2 - Session 2.0

Course 2

This course follows from the Practitioner Training Course and equips you with the foundational skills to design, build, and operate modern Mobile Phone Data (MPD) workflows for policy use.

- You will learn how to manage and process MPD datasets, integrate reference data, and design secure, privacy-conscious data pipelines.

- Through guided exercises and instructor-led sessions, you will gain hands-on experience in data quality assurance, developing core aggregates, and transforming them into policy-relevant estimates.

- By the end of the course, you will be able to apply best practices for producing accurate, timely, and reliable MPD-derived insights to inform decision-making in your country context.

Recap of fundamentals

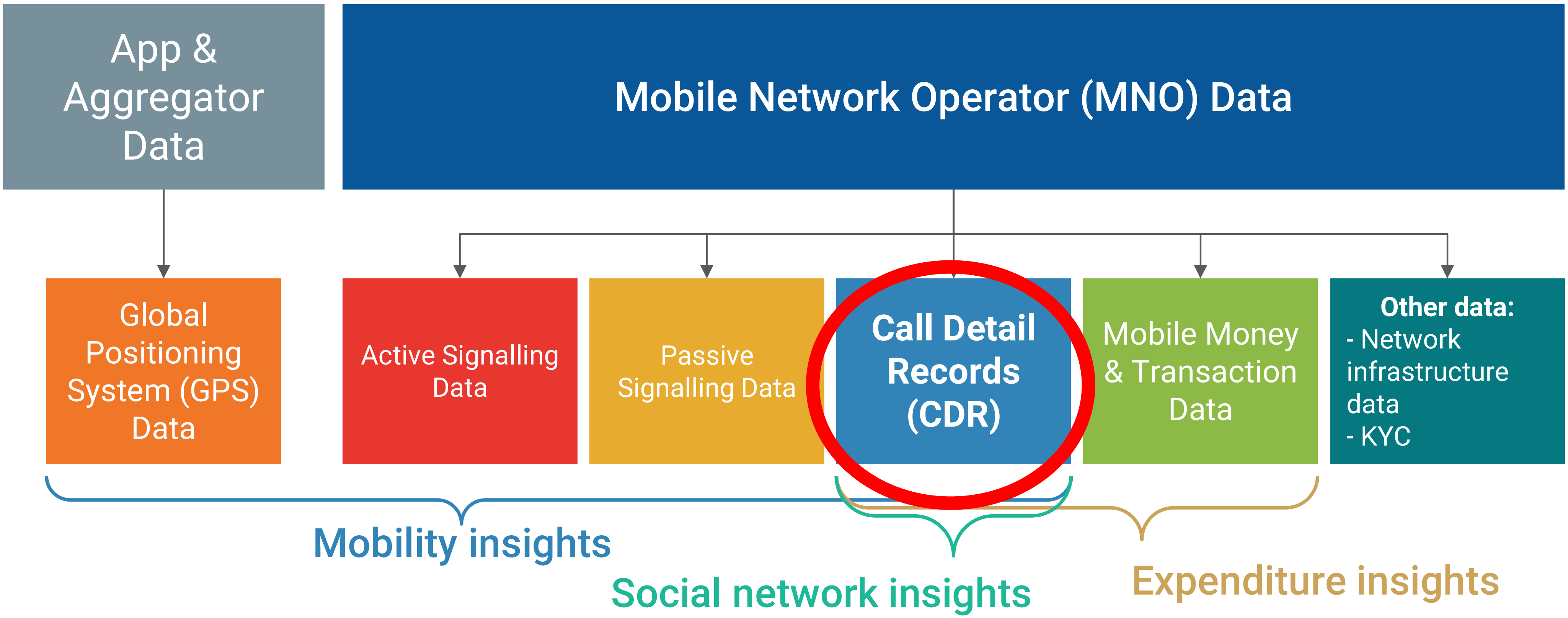
This short refresher session should help you get oriented before diving into the technical material.

It provides a quick overview of key concepts - data types, data pipelines, and data products to ensure everyone starts from a common baseline.

What do we mean by ‘mobile phone data’?

Mobile Phone Data (MPD)

This course covers the use of CDR records for data analysis.



Why CDRs?

This data is an opportunity to **address important data gaps** and/or **augment traditional data sources**

Advantages over other data types:

- Produced in **real-time**
- **Passively generated**, so doesn't require active data collection (and associated costs)
- **Routinely recorded** and **stored** by MNOs, happens automatically as per of service usage by subscribers
- Large geographic **scale** (often national-level)

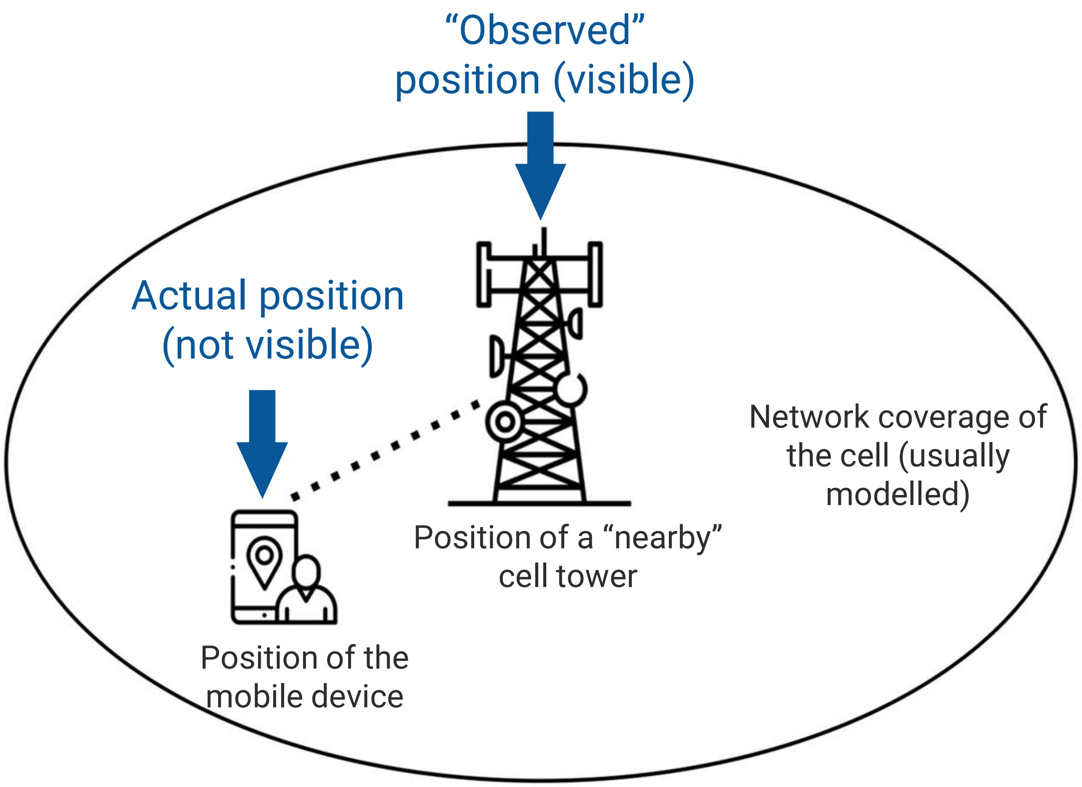
In low and middle income contexts, CDRs useful because:

- They are generated regardless of the type of mobile device
- Have higher penetration than smartphones

How CDR data is generated

CDR data

- Passively generated when a subscriber
  - Makes or receives **a call**
  - Sends or receives **an SMS**
  - Uses mobile **data**
- Location is at **the cell tower** level
- Routinely stored by MNOs for billing purposes



Key information contained in CDR data

MSISDN	CELL_ID	TIMESTAMP
AA204V1542DCA00	451154211	2016-10-10 15:35:25
AA20[illegible]	[illegible]312	20[illegible]3:45
AA204V[illegible]	[illegible]312	20[illegible]1:56
AA204V1542DCA03	451354312	2016-10-10 21:59:32
AA204V1542DCA04	452616792	2016-10-10 22:42:23
B45QHV45CAEVA5	476126941	2016-10-10 08:13:21
B45QHV45CAEVA6	476126941	2016-10-10 08:14:15
B45QHV45CAEVA7	47612[illegible]	[illegible]016-10-10 08:14:59
B45QHV45CAEVA8	47612[illegible]	[illegible]016-10-10 12:41:01
B45QHV45CAEVA9	476126941	2016-10-10 13:10:45
B45QHV45CAEVA10	476126941	2016-10-10 15:20:43

Calling party identifier (pseudonymised)

Timestamp

Cell ID (location)

Other variables in CDR data

- Receiving party identifier (pseudonymised)
- Network event type

CDR data do NOT contain information about the contents of a network event

Pseudonymization

- Pseudonymization means removing personally identifiable information by replacing it with random strings.
- The same original value always gets the same strings after pseudonymized

Example:
A phone number in mobile data is replaced with a random string like "A1B26s9Lrjd8UK7g45NC3".

Attribute	Original value	algorithm	Encrypted value
Phone No	654003453	SHA256	7cdab53309c015854b433f0f34d6cbc015104f9c42e36539b35ef974b26122d9
IMSI	611050105799949	SHA256	5c0c2c3675e41471d746fbe721e435e9ac4cdcba3406d78be89e5a87cab2d5f9
IMEI	865760021379230	SHA256	1341a37ec42db25140af0c6f7cc6e28b2b45f5d66832f6337fd6995e0d1f1582

Negotiating data access with Mobile Network Operators (MNOs)

Potential Incentives for MNOs

Indirect Benefits from Collaboration in Statistics

Regulatory compliance & legal mandates Legal mandate to clarify and de-risk data sharing	**Public Recognition & CSR** Boosting MNO corporate social responsibility profile.	**Data Reciprocity** Receive data from NSO in return.

Direct Benefits to Increased Capacity

New Products & Monetization Potential revenue stream outside of official statistics.	**Data Analysis Capacity Building** Utilize the expertise and infrastructure from the project.	**Innovation and R&D** Increase the quality of data analytics.

Levels of Access for NSO (Government)

Data User	**Level 1: NSO as a Data User and Disseminator only.** The NSO **lacks the legal mandate or technical capacity** for raw data processing. The MNOs or a service provider process the data according to the NSO's quality standards. The NSO's role is to **validate the final, aggregated outputs** and disseminate the statistics. This is the most common model when legal or technical hurdles are significant.	Example: Ghana
Data Processor	**Level 2: NSO as a Processor and Designer.** The NSO may not be the data controller but **has the technical capacity and trust** to access a pre-processed data stream from the MNOs or a technical service provider. The NSO can **design the processing algorithms and quality framework** and conduct final statistical validation before dissemination.	Example: Indonesia
Data Controller	**Level 3: NSO as Data Controller and Processor.** The NSO has the **legal mandate, technical capacity, and public trust** to receive raw, granular MPD directly from MNOs, process it into statistical outputs, and then disseminate it. This is the highest level of access.	Example: Oman, Estonia

Summary of Data Access Negotiations

1. It is important to **understand what incentives and disincentives Mobile Network Operators have to share data**. If clearly acknowledged, they can be strengthened or mitigated.
2. Various models have been successfully used to produce statistics with mobile network data, ranging from full to no raw data transfer. **The maturity of partners in the project determines, which data access model is preferred.**
3. Partnerships for data access are formalized in agreements, either through:
   a. Non-disclosure agreements (NDAs) to ensure confidentiality, even on the individual level.
   b. Memorandum of Understanding (MOU) to set up a general working relationship or describe data sharing in case of legal basis, and/or
   c. Data-sharing agreements (DSAs), which define all terms and conditions under which data sharing happens for a specific project.

Access the manual on accessing Mobile Phone Data from here: https://www.data4sdgs.org/roadmap-accessing-mobile-network-data-statistics

MPD policy applications

Many application opportunities

Mobile phone data (MPD) is a valuable source for the production of official statistics in a variety of policy domains where it is critical to provide reliable and up-to-date indicators on population's presence and mobility patterns.

Production of highly detailed, accurate statistics	**Fast and cost-effective collection of data**	**Increase the frequency of updated information**

The exploration of MPD

Many National Statistical Offices (NSOs) have launched activities aimed at using MPD. To support these efforts, the UN defines 6 areas for the exploitation of MPD:

- 1) Tourism statistics
- 2) Migration statistics
- 3) Census and dynamic population
- 4) Displacement in disaster context
- 5) Information society
- 6) Transport and commuting statistics

https://unstats.un.org/bigdata/task-teams/mobile-phone/index.cshtml UNBigData

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The exploration of MPD (summary)

Policy application	Measuring Sustainable Development Goals (SDG)	Dynamic Population Mapping	Migration & Disaster Management	Public Health & Epidemiology	Transport	Tourism	Socioeconomic Applications: Poverty Mapping
Indicators	- Proportion of individuals who used internet from any location in the previous three months - Proportion of the population covered by a mobile network	- Number of people present in each area of the territory at each moment in time	- Residential movements during a period of time - Temporal displacement of people during a period of time	- Population exposure to environmental health factors - Social contact	Origin-Destination matrices by transport mode, trip frequency and trip distribution	- Number of tourists (local and foreign) that visit each area during a period of time, number and location of overnight stays, duration of a stay in a specific area or tourist spot	- Spatial distribution of poverty (fusion of MPD with other data sources)
Use cases	Sociodemographic analysis of Internet access, Infrastructure and resource planning	Resident population mapping, census, event management Infrastructure and resource planning	Migration statistics Disaster and displacement statistics	Exposure to pollution, noise and heatwaves Disease spreading / epidemiology	Transport planning, modeling, concessions and management	Tourism statistics Crowd management Protection of natural and cultural heritage	Mean wealth index Accessibility to basic services

Assessing the impact of a given application

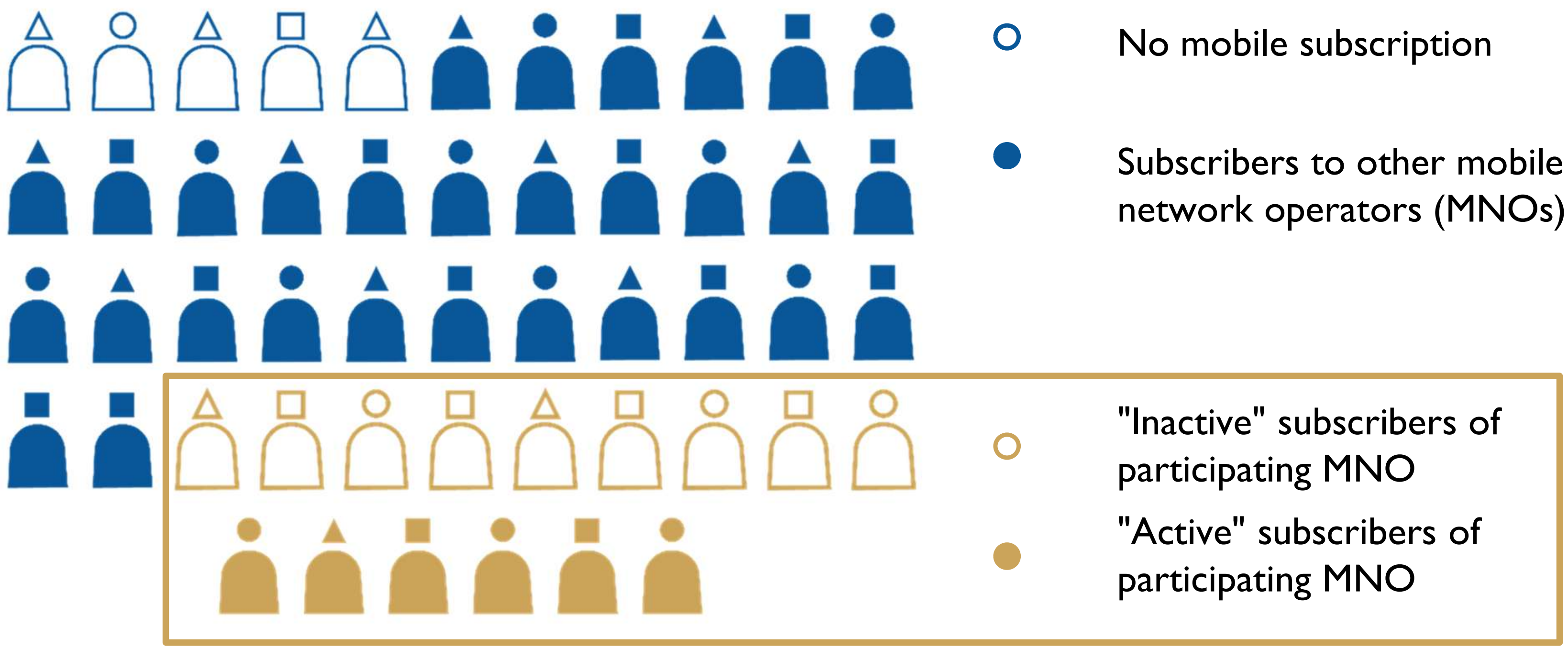
- ✓ Does it **address an existing gap** in official statistics?
- ✓ To what extent does it **reduce data collection costs** for a given indicator without compromising quality?
- ✓ To what extent does it **improve the quality** of a given indicator without increasing data collection costs?
- ✓ How effectively can the output indicators support governments in **making more efficient and informed decisions**?

Assessing the complexity of a given application

- ✓ What **characteristics must the input MPD** have?
- ✓ How complex are the required **MPD processing algorithms**?
- ✓ Does it need to be **combined with additional data sources**?
- ✓ What **level of accuracy is required** for the output indicators to be useful for decision-making?

Is mobile phone data ‘enough’?

Who is in the CDR data you are accessing?



Visual: Flowminder

Issue of bias

A whole population
you are trying to estimate

Visual: Flowminder

Examples of additional data for adjustment

Census: Gambia Bureau of Statistics (GBoS) included questions about ICT facility and device in the last Census conducted in 2023/24 (supported by the World Bank)

- Access to the mobile device
  - Mobile phone users or non-users
- Usage of the mobile services
  - Multiple-sim holding, phone sharing
  - MNO subscriptions
  - Expenditure on mobile phone services

Additional data sources are needed to scale subscriber level estimates to population scale estimates

Household Travel Survey

- Typical travel behavior on a weekday
- Home and work/education locations on the transport analysis zone (TAZ)

(Example) Data sources per country

Data sources	DRC	Ghana	Haiti
CDRs	Vodacom	Telecel (formerly Vodafone) Ghana	Digicel
Baseline population	WorldPop 2020: gridded population estimates (bottom-up)	Census 2021: sub-regional population estimates (GSS)	WorldPop, IHSI, UN Population Prospects: sub-regional population estimates
Secondary & "piggyback" surveys	Multi-Sector Needs Assessment (MSNA) 2023: update to weighting parameters (REACH)	Annual Household Income and Expenditures Survey (AHIES) 2022: phone users and non-users (GSS)	Multi-Sector Needs Assessment (MSNA) survey 2022: phone users and non-users (REACH)
Primary surveys	Micro-census 2021: covering 7 provinces, phone users and non-users (HH survey, led by Flowminder) Phone survey 2021: targeting phone users across the country from all MNOs (commissioned by Flowminder)	Phone survey 2022: targeting phone users across the country from all MNOs (commissioned by Flowminder, conducted by GSS)	Phone survey 2023: targeting phone users across the country from all MNOs (commissioned by Flowminder, conducted by Viamo)

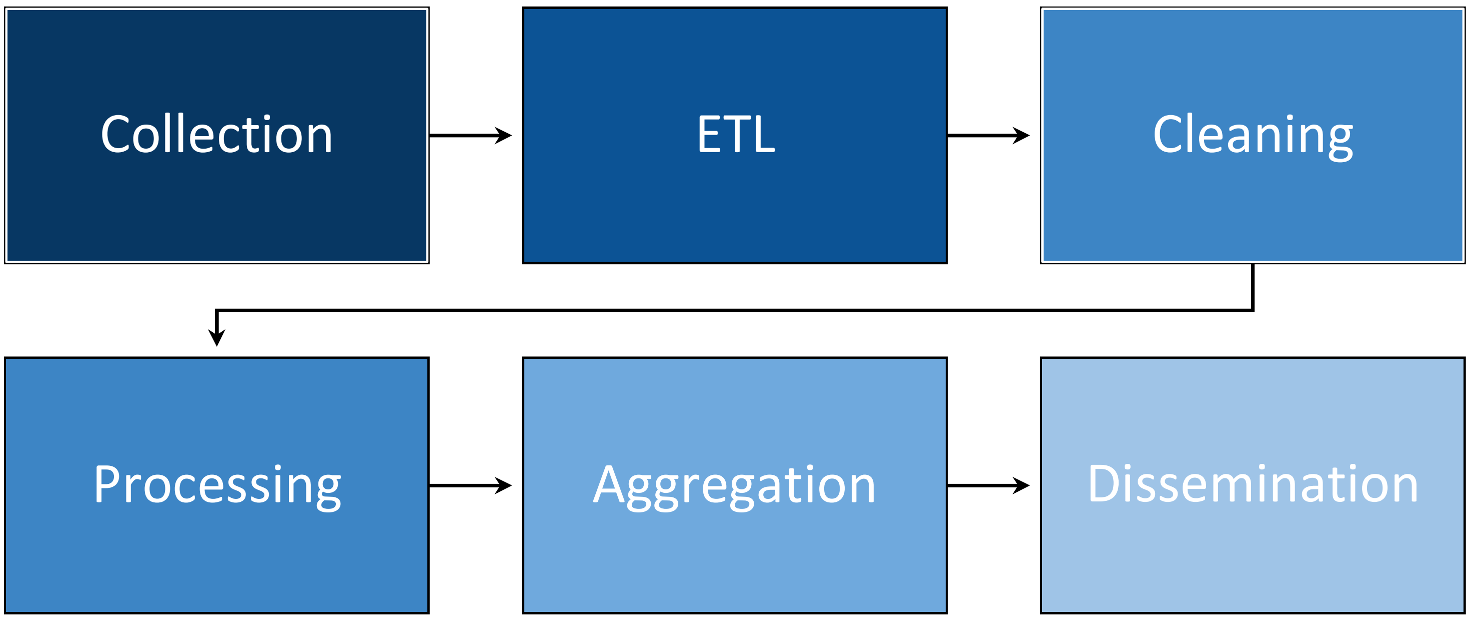
Source: Flowminder

Data pipelines

From Network Events to Statistics

Collection	ETL	Processing	Aggregation	Bias Adjustment and Scaling
Networks events (e.g. calls, sms, data sessions, location updates etc) occur on MNOs network MNO records these events for operational purposes (e.g. billing, network monitoring)	MNO pseudonymises the data Agreed data features are extracted and transformed into the necessary format Quality assurance checks (syntactic and semantic)	Data cleaning and filtering Data sampling Continuity models Detecting meaningful locations (home/work etc) Routing	Spatial and temporal aggregation based on domain specific methodology	Applied statistical methods to account for biases in the data Scale estimates from subscribers to population Quality assurance Statistical disclosure controls

Processing steps

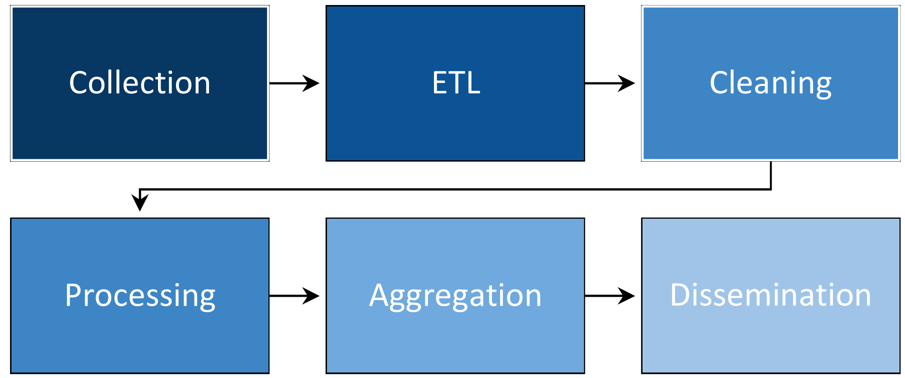


Data products

Dissemination

Several way to share resulting outputs:

- Reports
- Tables
- Maps
- Dashboards
- API-s



month	country	d_class	unique_visitors	stays_started	spent_nights	spent_days
01.09.2017	FI	1	6366	9447	0	9447
01.09.2017	FI	2	30735	42849	167672	205225
01.09.2017	FR	1	904	1001	0	1001
01.09.2017	FR	2	5704	8672	31783	39465
01.09.2017	GB	1	635	692	0	692
01.09.2017	GB	2	4355	5297	25144	29866
01.09.2017	GE	1	227	242	0	242
01.09.2017	GE	2	1230	1330	7160	8105
01.09.2017	IE	1	82	89	0	89
01.09.2017	IE	2	464	564	2708	3244
01.09.2017	LT	1	1185	1325	0	1325
01.09.2017	LT	2	7163	8789	16769	23788
01.09.2017	LU	1	74	102	0	102
01.09.2017	LU	2	316	414	1764	2145
01.09.2017	LV	1	8266	10083	0	10083
01.09.2017	LV	2	20518	24894	42843	62104

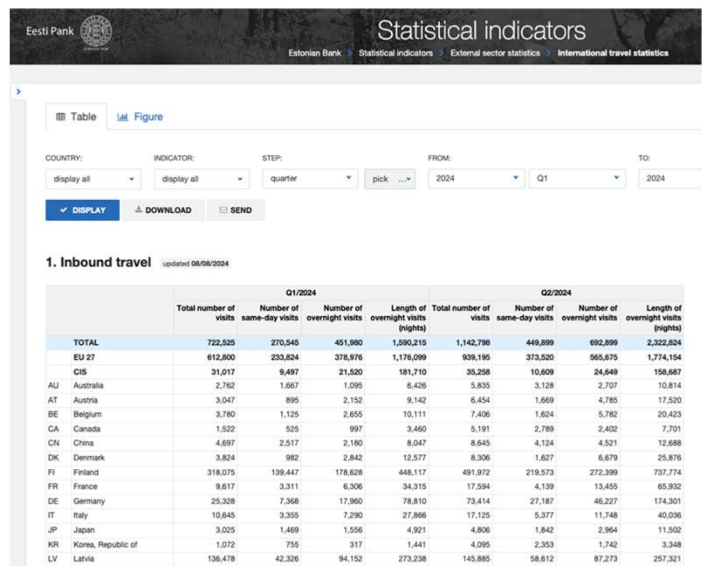

Eesti Pank
Statistical indicators
Estonian Bank > Statistical indicators > External sector statistics > International travel statistics

Table | Figure

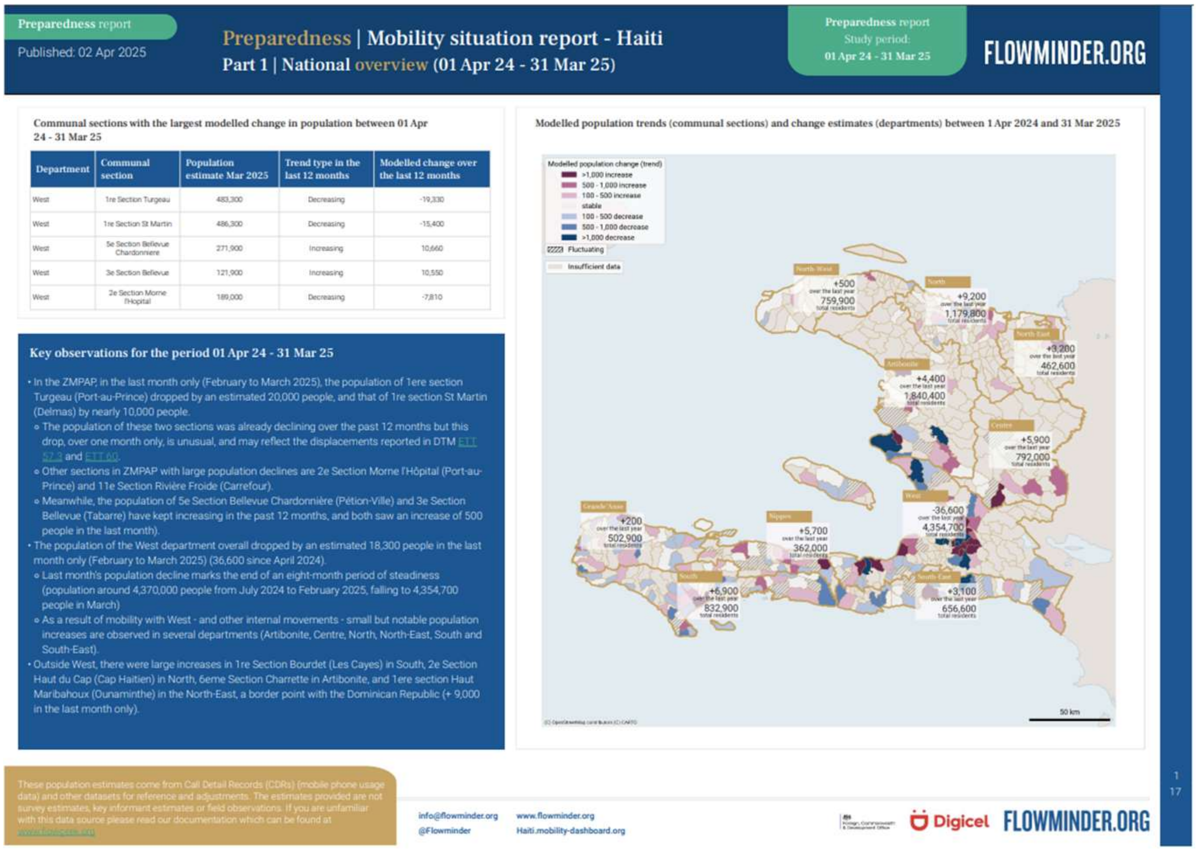
COUNTRY: display all | INDICATOR: display all | STEP: quarter | pick ... | FROM: 2024 | Q1 | TO: 2024

DISPLAY | DOWNLOAD | SEND

1. Inbound travel updated 08/08/2024

		Q1/2024				Q2/2024			
		Total number of visits	Number of same-day visits	Number of overnight visits	Length of overnight visits (nights)	Total number of visits	Number of same-day visits	Number of overnight visits	Length of overnight visits (nights)
	TOTAL	722,525	270,545	451,980	1,590,215	1,142,798	449,899	692,899	2,322,824
	EU 27	612,800	233,824	378,976	1,176,099	939,195	373,520	565,675	1,774,154
	CIS	31,017	9,497	21,520	181,710	35,258	10,609	24,649	158,687
AU	Australia	2,762	1,667	1,095	6,426	5,835	3,128	2,707	10,814
AT	Austria	3,047	895	2,152	9,142	6,454	1,669	4,785	17,520
BE	Belgium	3,780	1,125	2,655	10,111	7,406	1,624	5,782	20,423
CA	Canada	1,522	525	997	3,460	5,191	2,789	2,402	7,701
CN	China	4,697	2,517	2,180	8,047	8,645	4,124	4,521	12,688
DK	Denmark	3,824	982	2,842	12,577	8,306	1,627	6,679	25,876
FI	Finland	318,075	139,447	178,628	448,117	491,972	219,573	272,399	737,774
FR	France	9,617	3,311	6,306	34,315	17,594	4,139	13,455	65,932
DE	Germany	25,328	7,368	17,960	78,810	73,414	27,187	46,227	174,301
IT	Italy	10,645	3,355	7,290	27,866	17,125	5,377	11,748	40,036
JP	Japan	3,025	1,469	1,556	4,921	4,806	1,842	2,964	11,502
KR	Korea, Republic of	1,072	755	317	1,441	4,095	2,353	1,742	3,348
LV	Latvia	136,478	42,326	94,152	273,238	145,885	58,612	87,273	257,321

Dissemination report example



Preparedness report

Published: 02 Apr 2025

Preparedness | Mobility situation report - Haiti

Part 1 | National overview (01 Apr 24 - 31 Mar 25)

Preparedness report
Study period:
01 Apr 24 - 31 Mar 25

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Communal sections with the largest modelled change in population between 01 Apr 24 - 31 Mar 25

Department	Communal section	Population estimate Mar 2025	Trend type in the last 12 months	Modelled change over the last 12 months
West	1re Section Turgeau	483,300	Decreasing	-19,330
West	1re Section St Martin	486,300	Decreasing	-15,400
West	5e Section Bellevue Chardonniere	271,900	Increasing	10,660
West	3e Section Bellevue	121,900	Increasing	10,550
West	2e Section Morne l'Hopital	189,000	Decreasing	-7,810

Key observations for the period 01 Apr 24 - 31 Mar 25

- In the ZMPAP, in the last month only (February to March 2025), the population of 1ere section Turgeau (Port-au-Prince) dropped by an estimated 20,000 people, and that of 1re section St Martin (Delmas) by nearly 10,000 people.
  - The population of these two sections was already declining over the past 12 months but this drop, over one month only, is unusual, and may reflect the displacements reported in DTM ETT 57.3 and ETT 60.
  - Other sections in ZMPAP with large population declines are 2e Section Morne l'Hôpital (Port-au-Prince) and 11e Section Rivière Froide (Carrefour).
  - Meanwhile, the population of 5e Section Bellevue Chardonnière (Pétion-Ville) and 3e Section Bellevue (Tabarre) have kept increasing in the past 12 months, and both saw an increase of 500 people in the last month).
- The population of the West department overall dropped by an estimated 18,300 people in the last month only (February to March 2025) (36,600 since April 2024).
  - Last month's population decline marks the end of an eight-month period of steadiness (population around 4,370,000 people from July 2024 to February 2025, falling to 4,354,700 people in March)
  - As a result of mobility with West - and other internal movements - small but notable population increases are observed in several departments (Artibonite, Centre, North, North-East, South and South-East).
- Outside West, there were large increases in 1re Section Bourdet (Les Cayes) in South, 2e Section Haut du Cap (Cap Haitien) in North, 6eme Section Charrette in Artibonite, and 1ere section Haut Maribahoux (Ounaminthe) in the North-East, a border point with the Dominican Republic (+ 9,000 in the last month only).

Modelled population trends (communal sections) and change estimates (departments) between 1 Apr 2024 and 31 Mar 2025

These population estimates come from Call Detail Records (CDRs) (mobile phone usage data) and other datasets for reference and adjustments. The estimates provided are not survey estimates, key informant estimates or field observations. If you are unfamiliar with this data source please read our documentation which can be found at

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www.flowminder.org
Haiti.mobility-dashboard.org

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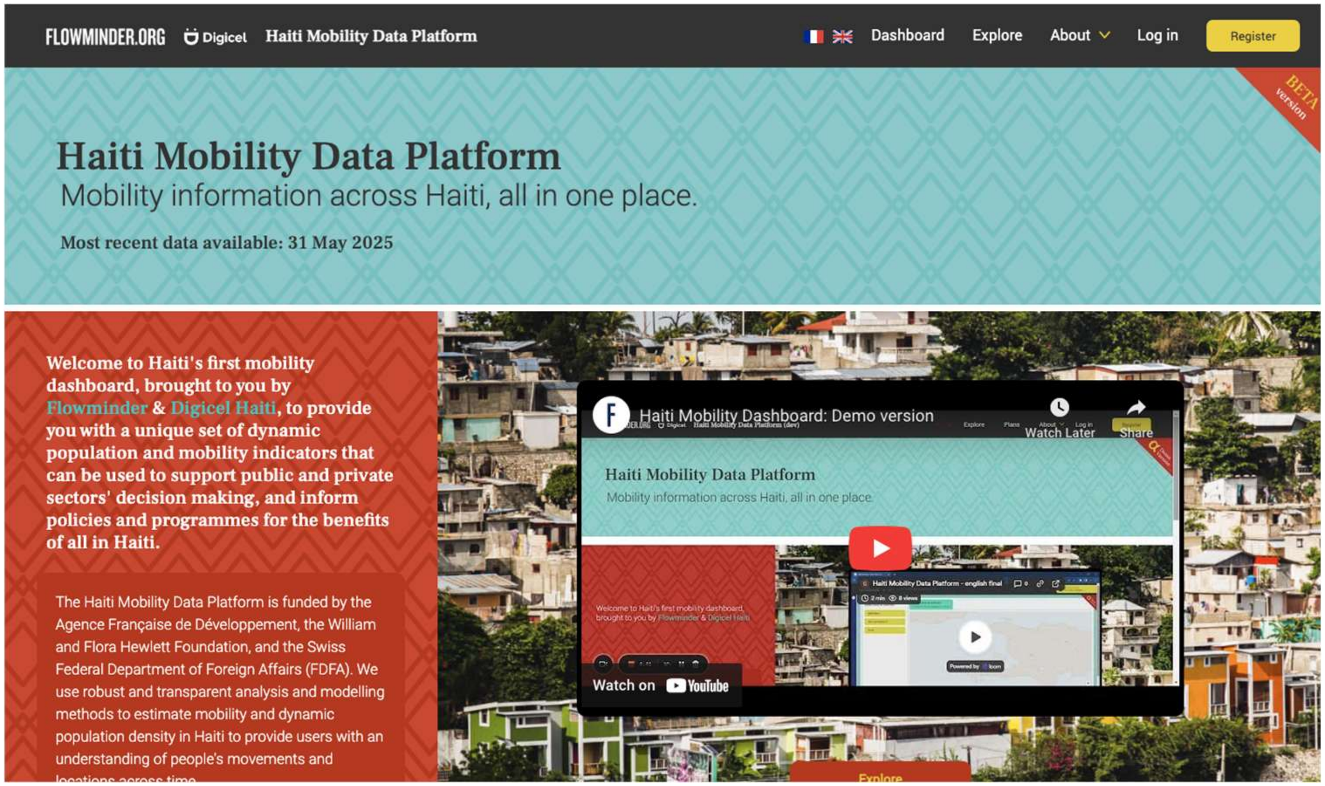
Dissemination data platform example



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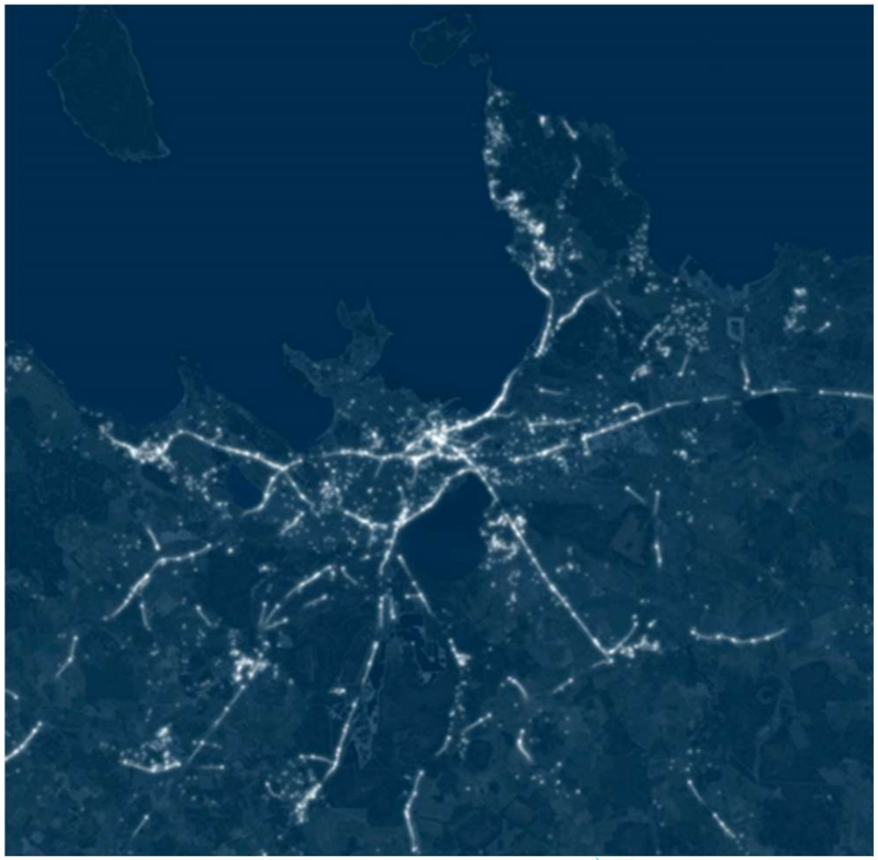
Dissemination animation example



The GDF-MPD window is a multi-partner window funded by the Spanish Government, Ministry of Economy, Commerce and Business. The GDF-MPD program is jointly managed by DEC (DECDG, DIME), Poverty and Digital Development Global Practices in technical partnership with the International Telecommunications Union (ITU).